

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

23rd May 2023

Subject SP4 – Pensions and Other Benefits

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
- 3. Attempt all questions beginning your answer to each question on a separate sheet.*
- 4. Mark allocations are shown in brackets.*
- 5. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** You are the actuarial advisor to the Government of a developing country. The country provides a basic level of State-provided pension to all citizens above a certain age. Over the last few years, the cost of the State-provided pension provision has been gradually rising.

At the same time, the Government feels that State-provided pension is not enough to generate an adequate post-retirement income and needs to be supplemented by private pension provision. Despite several initiatives undertaken by the Government over the last few years with respect to building awareness about private pension provision, the overall penetration of pensions continues to be poor in the country.

- i)** Give reasons as to why the cost of State-provided pension provision may be increasing and there is an overall lack of private provision in the country. (10)

After several rounds of discussions with you, the Government feels that there is a need to push pension provision more aggressively. It is now considering introducing a new National Pension System for the corporate sector, based on a Defined Contribution Pension system. Under this Scheme, an employee will automatically be enrolled in the scheme on joining a company's employment; but can decide to opt out.

Employees will contribute up to 10% of basic salary. Employers will be required to match employee contributions up to a limit, 5% of basic salary. Contributions will be accumulated in individual accounts till retirement and invested in a selection of funds chosen by the individual. The Scheme will be managed by the Government. At retirement, part of the balance will be paid as a lump sum and a minimum amount would need to be taken as an annuity to be bought out with an insurance company.

- ii)** Outline the advantages and disadvantages of the suggested proposal. (10)

To further encourage pension provision, the Government is looking to provide additional tax benefits to employer provided Defined Contribution schemes, the prevalence of which has been historically very low. The Government anticipates that due to this move, there will be a significant increase in employer provided Defined Contribution schemes over the next few years and is therefore, looking to introduce legislation in respect of the provision of Defined Contribution schemes.

- iii)** Set out the areas where regulations may be imposed for Defined Contribution Schemes. (10)
[30]

- Q. 2)** A large country has limited universal social security and pensions are provided primarily by way of employer sponsored schemes. The schemes can be funded separately through a Trust or set up as unfunded schemes through book provision in the company's accounts.

While the pensions market has been well-regulated over the years, the Government is worried about the overall security of members.

They have asked you for your views on the following proposals:

- a) Require adequate disclosure to scheme members of the funding and investment strategy as well as overall operations of the scheme
- b) Prescribe a pattern of investments that pension funds can make, including a minimum requirement to hold at least 40% in Government bonds

- c) Require all pension schemes to contribute to a central insolvency fund
- d) Set out minimum funding requirements for pension funds

- i) Discuss disadvantages of unfunded schemes and issues in moving to a funded approach. (4)
 - ii) Discuss the rationale for each of the above proposals. (4)
 - iii) Set out the advantages and disadvantages of the above proposals as a means of providing extra security to members. (12)
- [20]**

Q. 3) A medium size pension scheme had their actuarial valuation done three years ago. During the covid, the scheme has experienced some deaths. The valuation is due; however, the trustees have asked you to carry out a mortality analysis to consider the impact.

- i) Outline the key steps in using membership data to carry out an analysis of experience of mortality in a pension scheme. (6)

A trustee of other similar scheme has asked you to use the results of above analysis in his scheme in order to avoid duplicity of work.

- ii) Discuss the appropriateness of using the results from such an analysis for a scheme with 50 members, for using in a valuation. (6)
- iii) Outline what adjustments needs to be given in this situation. (2)

The previous funding valuation for the above medium size scheme in part (i) above assumed that all members converted the maximum possible amount of pension which is two-third of the total amount into a cash lump sum at retirement.

Similar analysis, as above, for the take up rate was also carried out and it was observed that 90% of members retiring in the last 6 years took the maximum possible cash lump sum at retirement, with the remaining 10% of members taking no cash lump sum at retirement.

- iv) Describe the points the Actuary should consider in deciding if it is appropriate to amend the assumptions for the funding valuation based on this analysis. (6)

The trustees are concerned about the security of benefits of members as maximum members opt out for the cash lump sum and it has been suggested that members should be required to obtain the consent of the scheme managers before exercising the option to convert pension to cash at retirement.

- v) Discuss the advantages and disadvantages of this suggestion. (5)
- [25]**

Q. 4) The Pension Authority of a Country has appointed retirement advisors. Their role includes but not limited to advising the benefits of pension schemes regulated by Authority and supporting them in making simple decisions based on the profile of client, current & future liabilities, investments allocation and selection of Pension Funds.

A wealthy mid-aged individual has a large cash lump sum. He has asked you who is a retirement advisor and an Actuary to advise on how he might invest this sum to provide for himself and his family in retirement and to be able to leave an amount as an inheritance.

- i) Describe how the retirement advisor who is an Actuary also would assess the individual's future liabilities. (4)

The individual has already had enough protection cover in case his untimely death; however, he is more concerned of his being living more than expected in absence of regular income in future when he is not able to work.

- ii) Identify the key risks from the perspective of an individual, which you as an Actuary should consider when giving them advice. (2)

- iii) Describe how each of the risks in (ii) might be mitigated. (6)
[12]

- Q. 5)** A pension scheme has been facing an issue of funding from their sponsors as the last year's valuation showed a surplus and hence a nil contribution was paid by them and this year's valuation has showed a deficit in the pension scheme while no major event has happened during the year.

The trustee of the scheme confirmed that this was the first-time during last year's valuation when the assets were valued at the market value.

- i) Discuss the advantages and disadvantages of using market values for the valuation of the assets of a pension scheme. (4)

The Actuary has suggested to use the discounted cash flow approach instead.

- ii) Suggest discounted cash flow values may not be suitable alternatives to market values for the valuation of assets of a pension scheme. (3)

It is three years since the last scheme valuation. The average annual return on the domestic equity portfolio of a pension scheme over the three years has been lower than half of the share market return over the ten years prior to the last scheme valuation.

- iii) Suggest possible reasons for this relative fall in returns while considering both market and specific factors of the scheme. (6)
[13]
